

Improving Preparedness of Communities for Evacuations using ZEVs

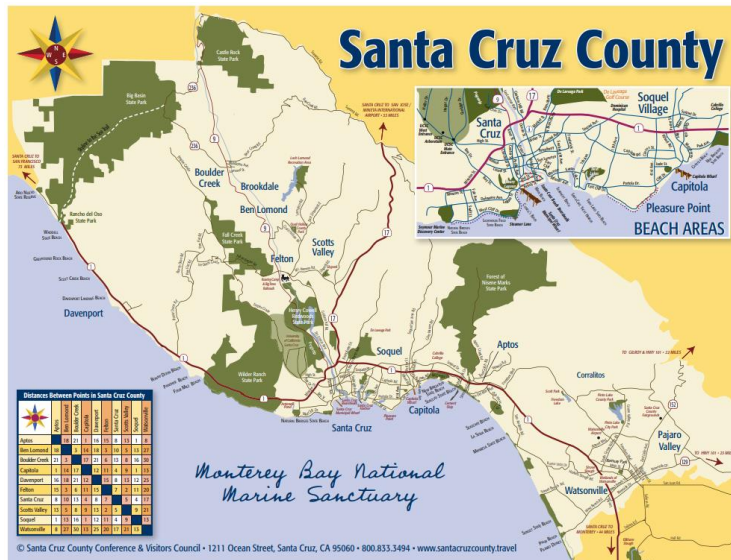
Development of ZEV Evacuation Readiness
Score - II

Presenter: Osman Saleem

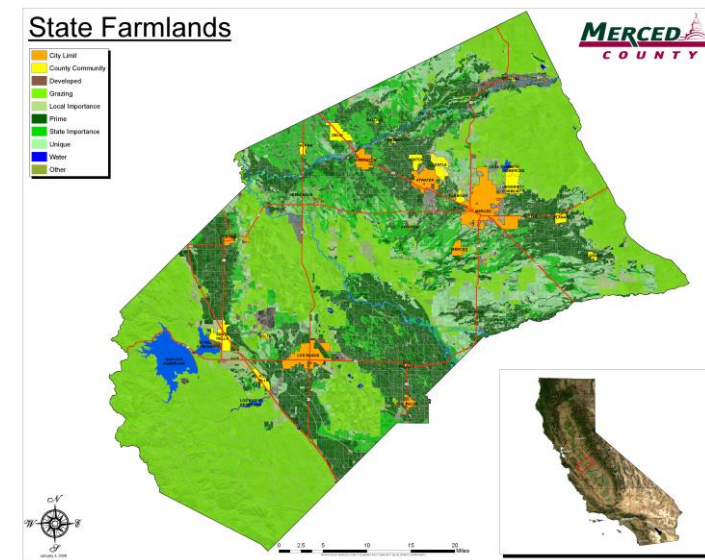
December 5th, 2024

Intro to ZEV Evacuation Readiness Score

- ZEV evacuation readiness score provides a basis to categorize each county based on its preparedness to successfully carry out a zero-emission vehicle evacuation.

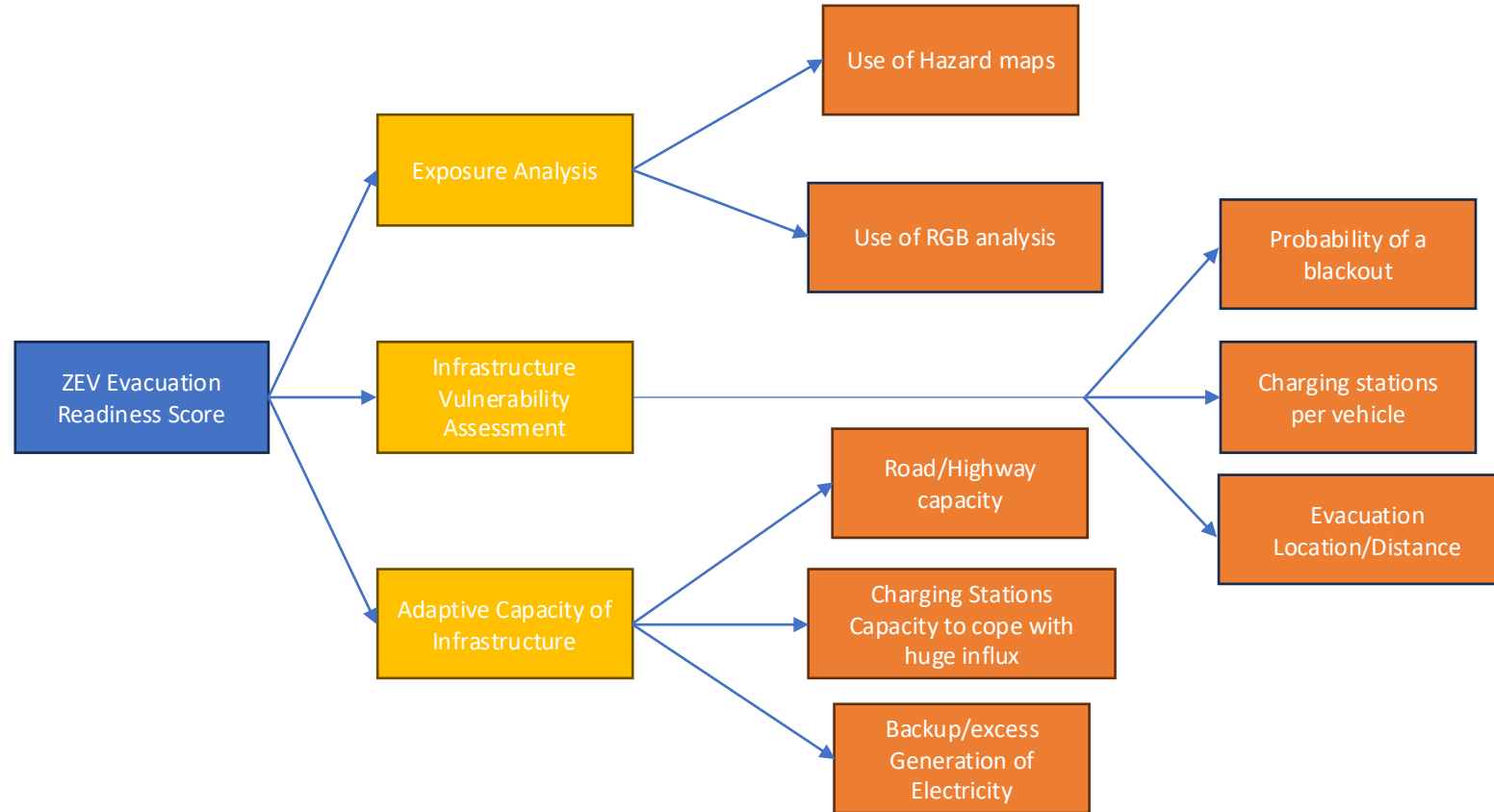


ZEV Score: ★★★★★



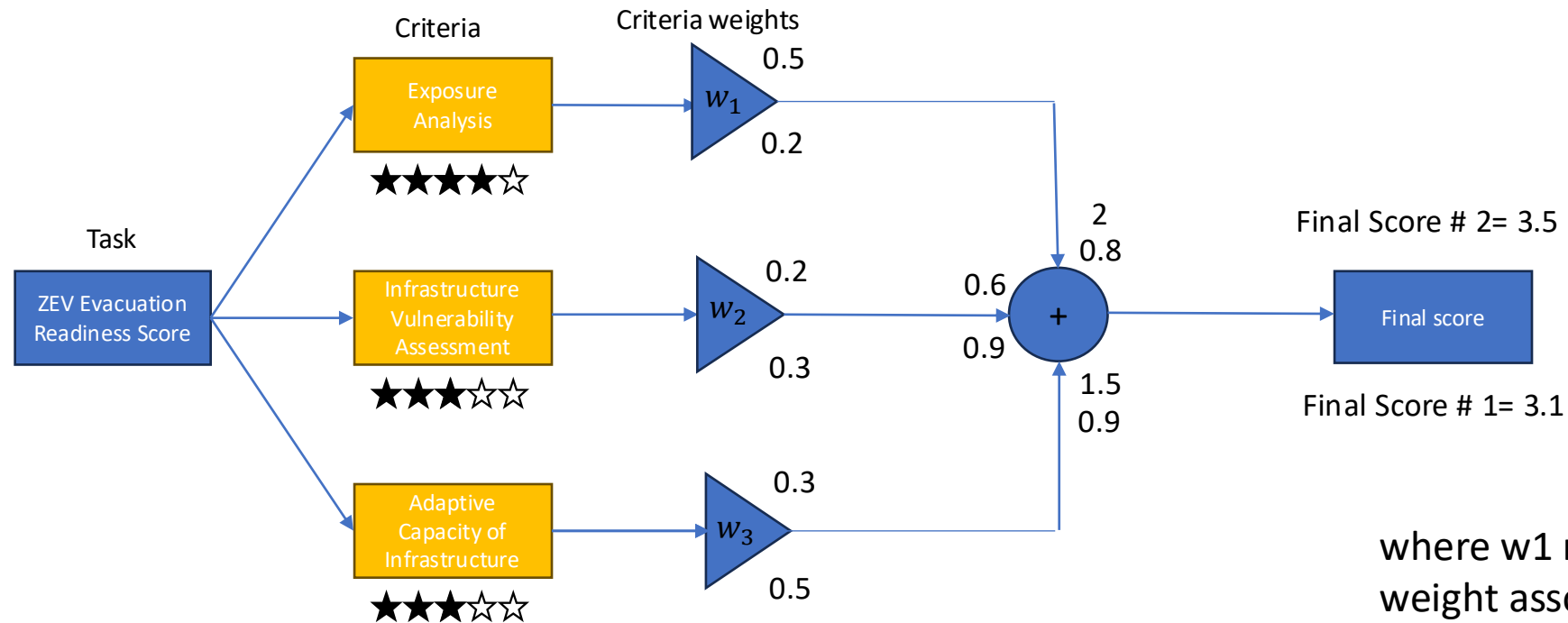
ZEV Score: ★★★★★

ZEV Evacuation Readiness Score Framework



Now, lets introduce the concept of weights!

ZEV Evacuation Readiness Score Weights Selection



$$R(EI, IVS, ACIS) = EI \cdot w_1 + IVS \cdot w_2 + ACIS \cdot w_3$$

where w_1 represents the weight associated with the EI analysis, w_2 represents the weight associated with the IVS analysis, w_3 represents the weight associated with the ACIS analysis.

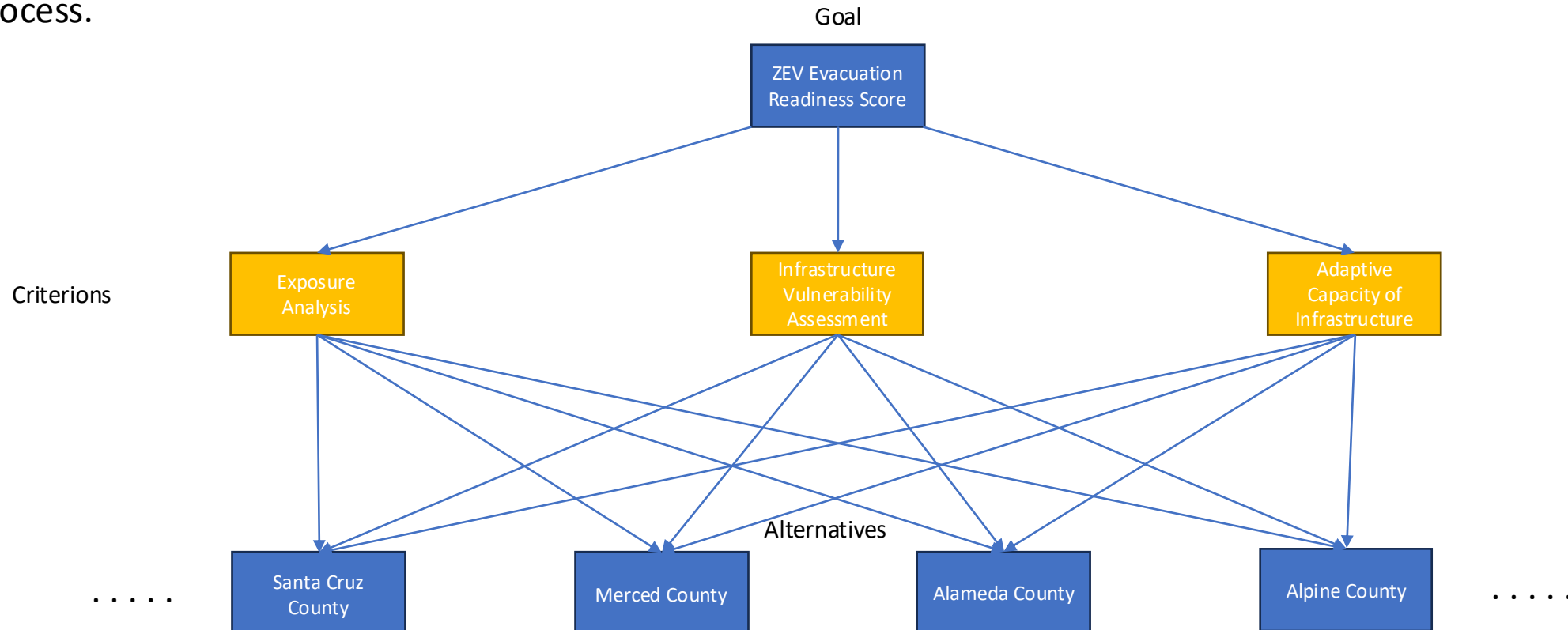
But to present ZEV score 2.0, with new weights, we introduce a concept of fuzzification!

Fuzzification of Scoring Methodology

Why Fuzzification:

- Dealing with uncertainty
- Handling vagueness and ambiguity
- Improves decision making
- Help analyze real-world scenario better

Previously, we were using simple “Analytical Hierarchy Process”. But for ZEV Score 2.0 we will be using Fuzzification process.



Weights Comparison

Weights calculated using Fuzzified AHP

$$w_1(\text{for Exposure Analysis}) = 0.24$$

$$w_2(\text{for Infrastructure Vulnerability Analysis}) = 0.19$$

$$w_3(\text{for Adaptive Capacity Infrastructure}) = 0.668$$

Weights calculated using simple AHP

$$w_1(\text{for Exposure Analysis}) = 0.15$$

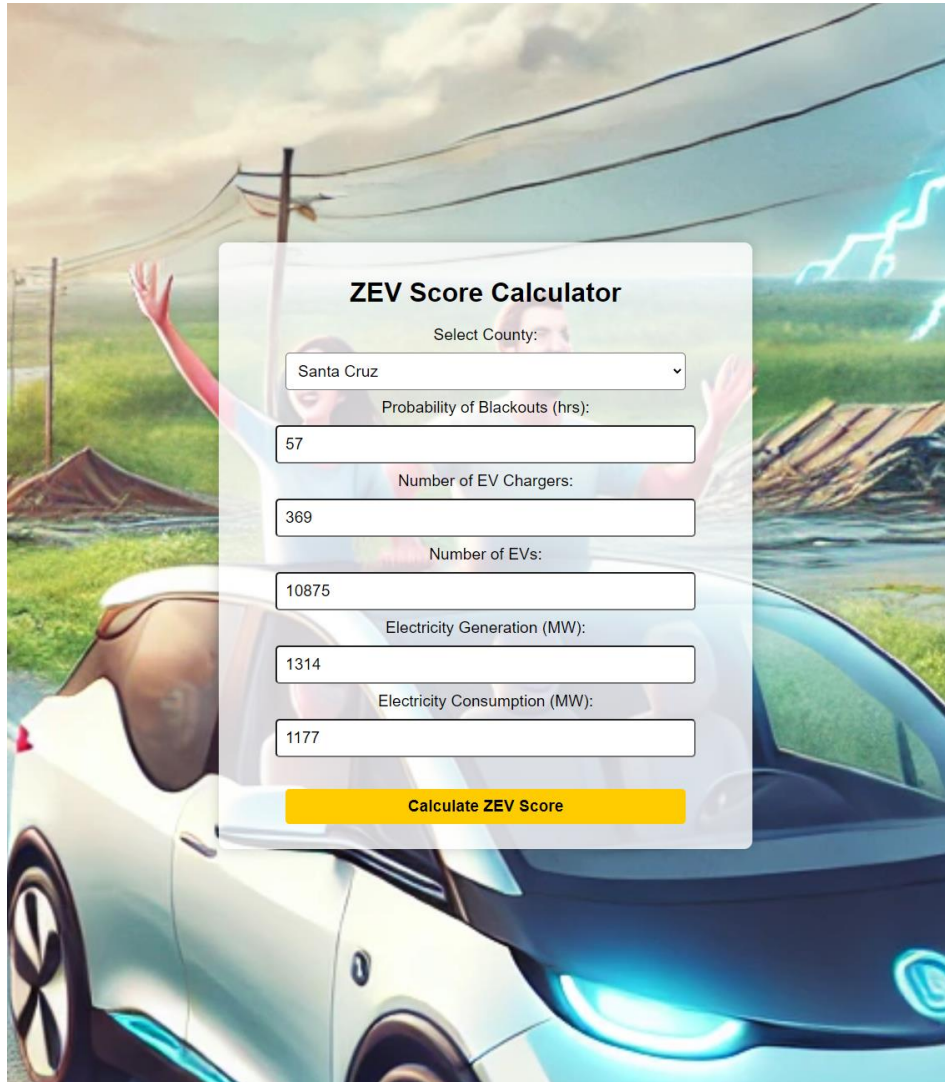
$$w_2(\text{for Infrastructure Vulnerability Analysis}) = 0.2$$

$$w_3(\text{for Adaptive Capacity Infrastructure}) = 0.65$$

Fuzzified Analytical Hierarchy Process (FAHP) enhances the standard Analytical Hierarchy Process (AHP) by incorporating fuzziness, which provides a more nuanced and accurate decision-making framework.

Creation of ZEV Calculation Tool

Landing Page



The landing page features a background illustration of a white electric car with glowing blue headlights in the foreground. In the background, a woman with long dark hair and a pink shirt stands with her arms raised in a gesture of surprise or excitement. Behind her, a green field is visible under a cloudy sky with a bright blue lightning bolt striking down. A wooden structure, possibly a fence or part of a building, is partially visible on the right. A semi-transparent white box with a light blue border is centered on the page, containing the 'ZEV Score Calculator' form.

ZEV Score Calculator

Select County:
Santa Cruz

Probability of Blackouts (hrs):
57

Number of EV Chargers:
369

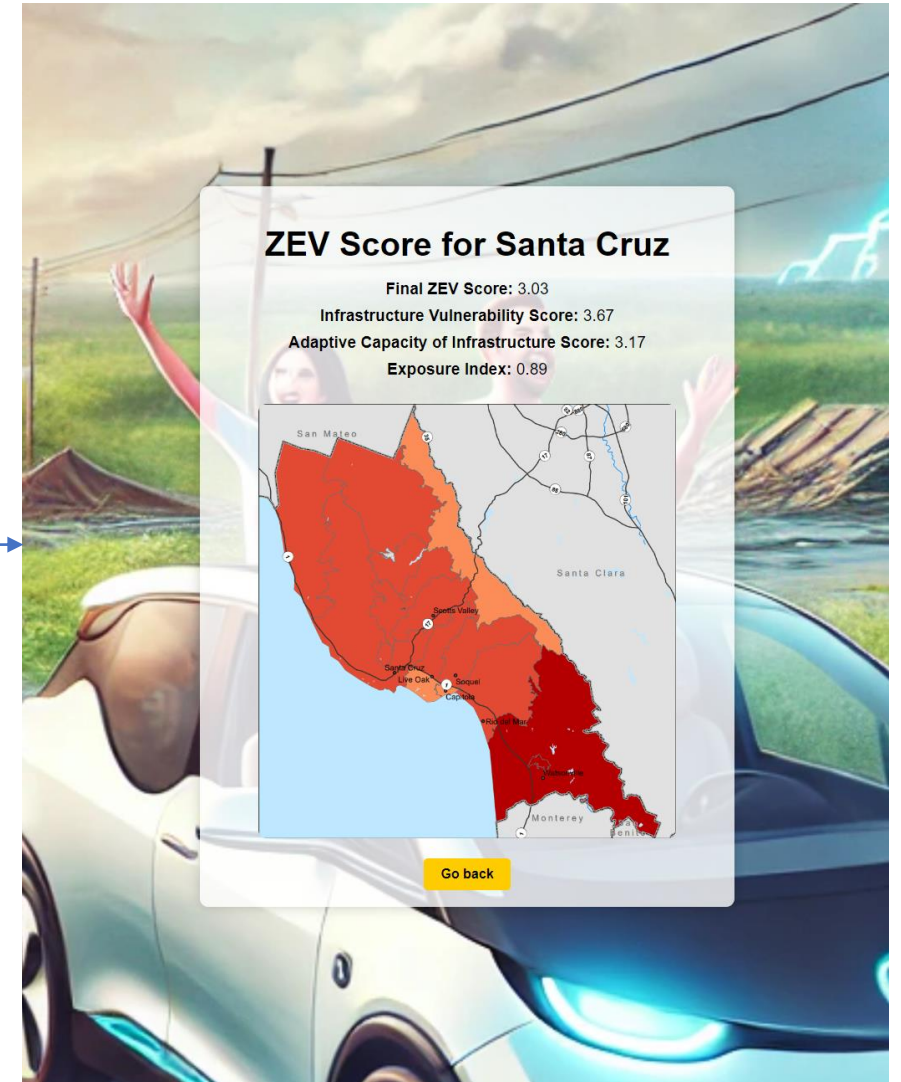
Number of EVs:
10875

Electricity Generation (MW):
1314

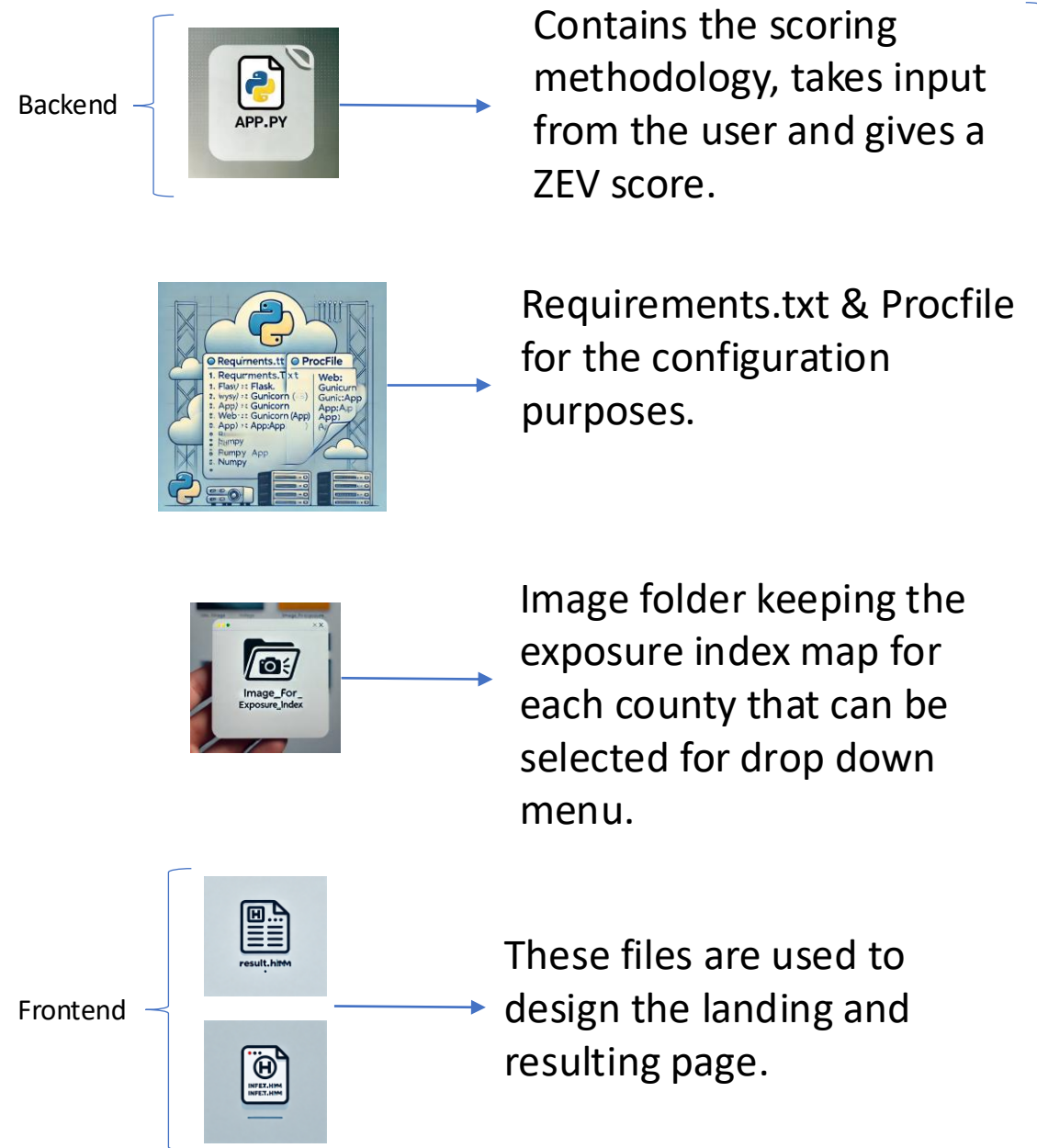
Electricity Consumption (MW):
1177

[Calculate ZEV Score](#)

Result Page



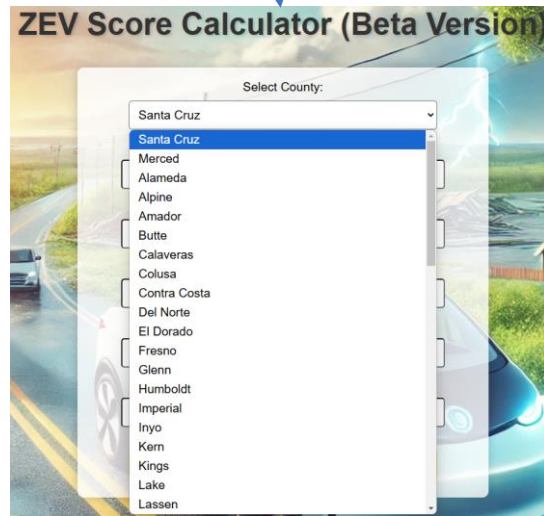
ZEV Score Calculation Web Page Development



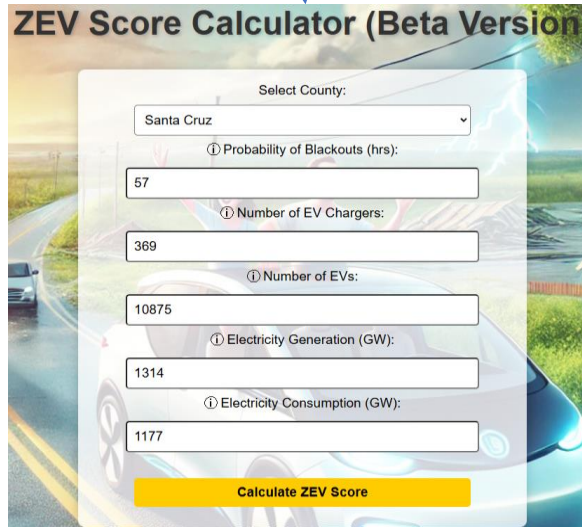
The image displays the ZEV Score Calculator web application interface. It features a form titled "ZEV Score Calculator" with a dropdown menu for "Select County:" set to "Santa Cruz". Below this are input fields for "Probability of Blackouts (hrs):" (57), "Number of EV Chargers:" (369), "Number of EVs:" (10876), "Electricity Generation (MW):" (1314), and "Electricity Consumption (MW):" (1177). A yellow "Calculate ZEV Score" button is at the bottom of the form. To the right, a summary panel titled "ZEV Score for Santa Cruz" shows the "ZEV Evacuation Readiness Score: 3.03". Below this, a flowchart breaks down the score into components: "Exposure Analysis: 0.89", "Infrastructure Vulnerability Assessment: 1.67", and "Adaptive Capacity of Infrastructure: 3.17". Further sub-components include "Probability of Blackout: 0.89", "EV to EV Charger Ratio: 0.89", "Evacuation Location & Distance: 3.08", "Highway Capacity: 4.00", "EV to EV Charger Ratio: 3.00", and "Back-up Power: 2.00". The final scores are: "Final ZEV Score: 3.03", "Infrastructure Vulnerability Score: 3.67", "Adaptive Capacity of Infrastructure Score: 3.17", and "Exposure Index: 0.89". A map of Santa Cruz is shown at the bottom of the summary panel, with a "Go Back" button.

ZEV Score Calculation Web Page Key Features

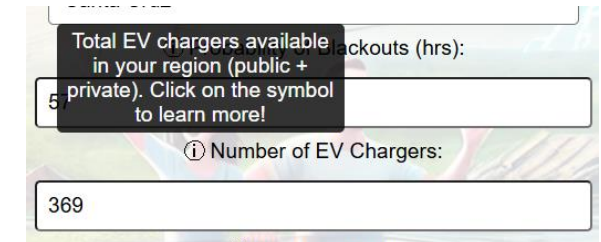
Contains every county in the drop-down menu.



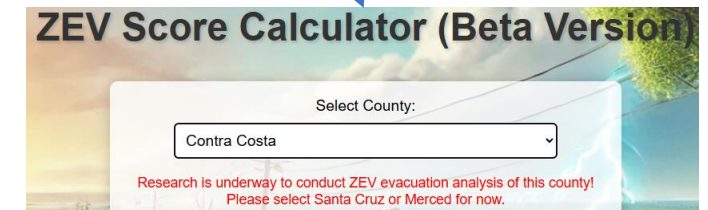
Whenever a County is selected, criteria values will automatically be adjusted for that county.



There is an information symbol right beside each criteria which can redirect you to further resources.

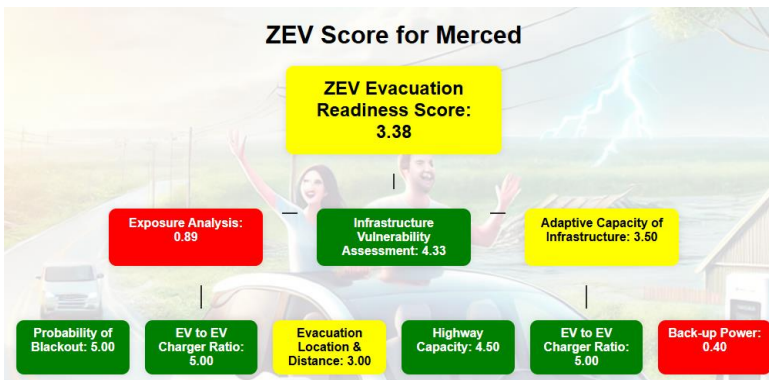
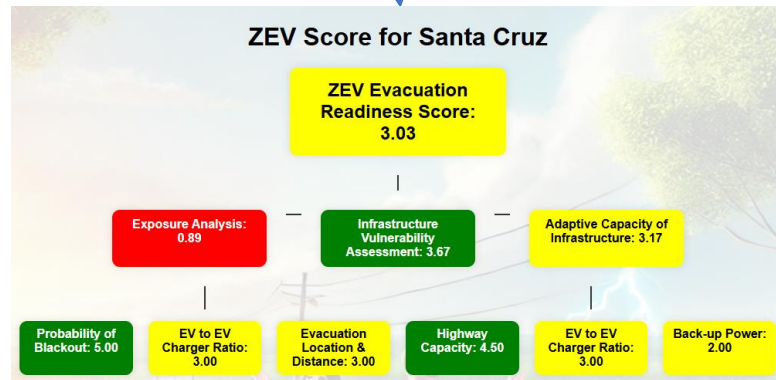


Right now, website is not optimized for counties except for Santa Cruz and Merced!

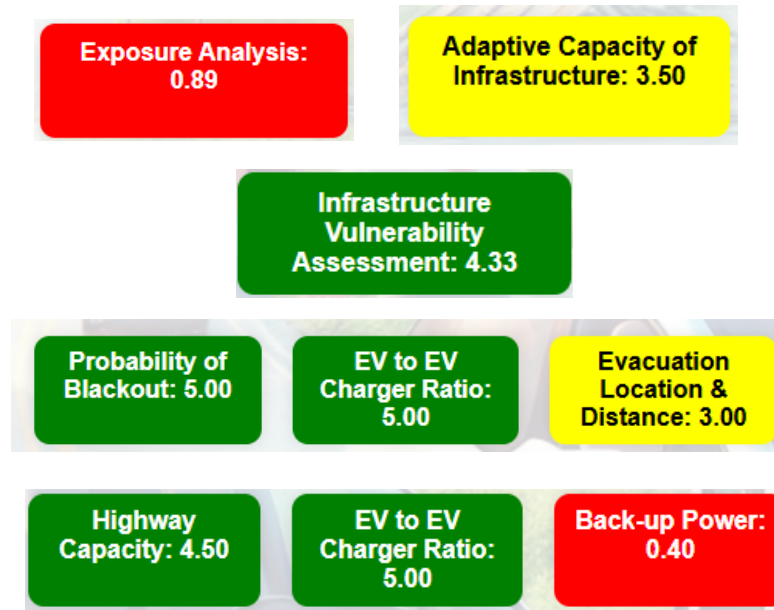


ZEV Score Calculation Web Page Key Features

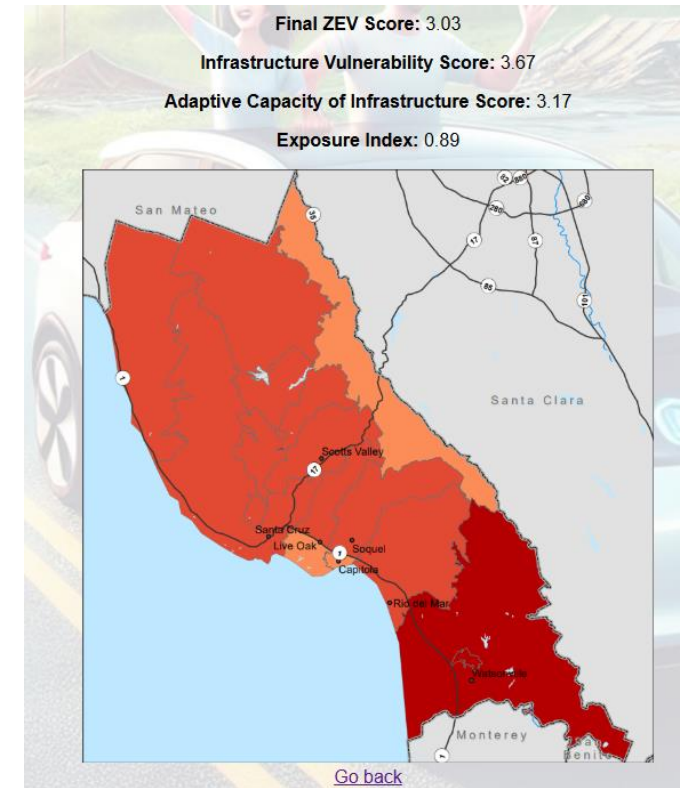
Output is represented in a flowchart for easier understanding of key factors enhancing or reducing the ZEV score.



Color coordination is introduced to better highlight the criteria/sub-criteria that need attention/improvement.



When exposure analysis is being carried out in runtime, the map being used is also displayed at the result page.



Link to the Website!



<https://zevscorecalculator-6a334960ec96.herokuapp.com/>

Thank you!

